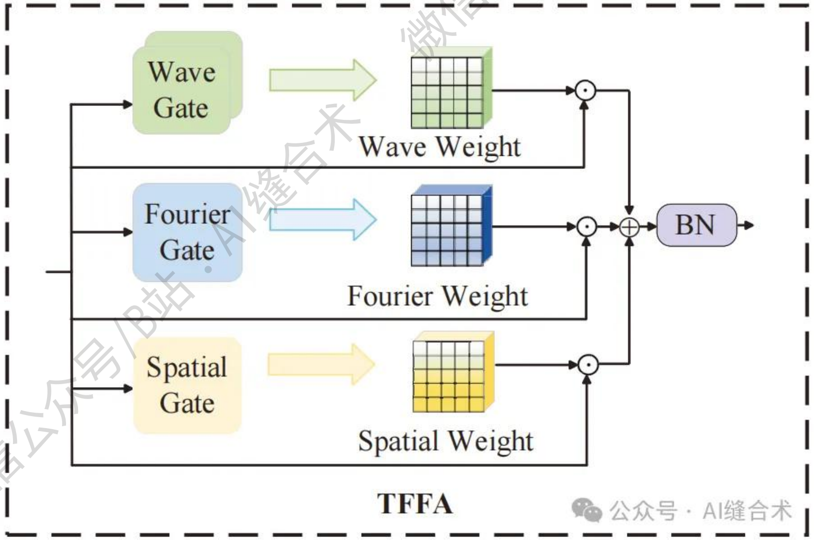
# 核心速览




**论文题目：** Decoding with Structured Awareness: Integrating Directional, Frequency-Spatial, and Structural Attention for Medical Image Segmentation

**中文题目：** 基于结构化感知的解码：整合方向性、频域-空间及结构注意力进行医学图像分割

**论文链接：** https://arxiv.org/pdf/2512.05494

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**核心速览：** 本文提出一种新型医学图像分割解码器框架，通过集成三个核心模块解决 Transformer 解码器在捕捉边缘细节、识别局部纹理和建模空间连续性方面的局限性，以提升分割准确性和模型泛化能力。

**论文内容详细解读:**

https://mp.weixin.qq.com/s/NBRgt9Rq1JFXJ38VAem92A

```
TFFA(
  (dog_conv): Conv2d(32, 16, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))
  (mexican_conv): Conv2d(32, 16, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))
  (wavelet_norm): BatchNorm2d(32, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
  (fourier_conv): Conv2d(64, 32, kernel_size=(1, 1), stride=(1, 1))
  (fourier_norm): BatchNorm2d(32, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
  (spatial_conv): Conv2d(32, 32, kernel_size=(1, 1), stride=(1, 1))
  (spatial_norm): BatchNorm2d(32, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
  (attention): Sequential(
    (0): Conv2d(96, 32, kernel_size=(1, 1), stride=(1, 1))
    (1): Sigmoid()
  )
  (final_conv): Conv2d(32, 32, kernel_size=(1, 1), stride=(1, 1))
  (act): GELU(approximate='none')
)
Input shape: torch.Size([2, 32, 256, 256])
Output shape: torch.Size([2, 32, 256, 256])
```